SE444: Introduction to Artificial Intelligence

Final Exam — Answer Sheet — Fall 2025

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Date: Monday, December 22, 2025 Time: 12:30 PM – 2:30 PM (120 min)

Instructions

- This is a closed-book exam. The University's code of ethics applies.
- Answer each exam question to the best of your knowledge.
- Show your work – any sign of serious effort will be considered.

Student Information

Student ID:	Student Name:

Part I. Critical-Thinking MCQ (6 pts, 30 min)

Instructions: Mark **X** for correct answer(s). Each question may have **one or more correct answers**.

Opt	Q1	Q2	Q3	Q4	Q5	Q6	Topics
A							Q1: d-Sep, Q2: Bayes
B							Q3: MDP, Q4: NE
C							Q5: STRIPS
D							Q6: Logic Agent
E							

Part II. Robot Navigation Under Uncertainty (16 pts, 60 min)

Q II.1: BN for Robot Perception (4 pts)

(a) Draw BN structure with 6 nodes [2 pts]: (b) $L \perp A \mid P$**? d-Sep [1 pt]: (c) Colliders + Explaining Away [1 pt]:**

Q II.2: Multi-Sensor Fusion (4 pts)

(a) $P(\textbf{Obstacle} = T \mid \textbf{Camera} = \textbf{Detect})$ **[2 pts]:**

(b) $P(\textbf{Obstacle} = T \mid \textbf{Cam}, \textbf{Lidar})$ **[2 pts]:**

Q II.3: Sensor-Based Navigation Decision (5 pts)

(a) $P(\textbf{Obstacle} = T \mid \textbf{Alert})$ **[2 pts]:**

(b) EU(Cross), EU(Wait), MEU-optimal action [3 pts]:

Q II.4: Project Code — Bayesian Reasoning (3 pts)

(a) Trace `compute_evidence()` **[1 pt]:** **(b) Trace** `bayes_update()`**, posterior [1 pt]:** **(c) If evidence=0.5? [1 pt]:**

Part III. Multi-Robot Cooperation (8 pts, 30 min)

Q III.1: Robot Task Planning — STRIPS (3 pts)

(a) STRIPS action `Pickup(robot, victim, loc)` **[1 pt]:**
Preconditions:

Add List:

Delete List:

(b) Forward search plan (MoveTo, Pickup, PutDown) [2 pts]:

Q III.2: Robot Coordination Game (3 pts)

(a) Strictly dominated strategies [1 pt]:

(b) Pure-strategy Nash Equilibria [2 pts]:

Q III.3: Project Code — Logic Agent (2 pts)

(a) `kb.tell()` **vs** `kb.ask()` **[1 pt]:**

(b) Trace for-loop, `kb.ask()` **results [1 pt]:**

Direction	North	East	South	West
`kb.ask()` returns				

Final `safe_moves` **=**